

# Phase-Based Motion Estimation: Benchmark of Two Reference Methods

Single-scale Fleet-Jepson vs. multi-scale coarse-to-fine (M-PME)

Purpose. Establish baseline benchmarks for dense phase-based displacement estimation, to be compared against the ImplicitKalman approach. Two standalone tools are evaluated on the same input pair (relief0 -> relief0\_sc1), both producing an identical six-panel diagnostic layout.

## Key finding

A single-scale phase estimate is fundamentally bounded to displacements below half a filter wavelength (the  $+\pi$  phase-wrap limit). It fails on the large / peripheral displacements produced by scaling. Decomposing the motion across a Gaussian pyramid (coarse-to-fine with rewarping) removes that ceiling: the wrap limit then applies only to the small per-level residual. On the test pair the multi-scale method reduces the warp reconstruction error by 97%.

| Method                      | Warp residual RMSE      | Verdict      |
|-----------------------------|-------------------------|--------------|
| Single-scale (Fleet-Jepson) | chaotic / aliased       | insufficient |
| Multi-scale (M-PME)         | 121.4 -> 4.11 px (-97%) | converges    |

Caveat for 2D->3D reconstruction: even the converging multi-scale field is too coarse for depth reconstruction. Phase information exists only at textured edges and is interpolated across flat regions, so the field is not sub-pixel smooth. This is the gap ImplicitKalman aims to close.

# Why phase encodes motion — and where it breaks

Fourier shift theorem. A local displacement  $\delta$  shifts the phase of a complex band-pass (Gabor / steerable) response linearly:

$$\Delta\phi = 2\pi k\delta$$

where  $k$  is the local spatial frequency (cycles/px). Inverting gives the displacement from the measured phase difference:

$$\delta = -\Delta\phi / k_{\text{local}}$$

Crucial detail (Fleet & Jepson):  $k$  must be the MEASURED instantaneous frequency  $k = \text{grad}(\phi)$ , estimated via  $\text{Im}(\text{grad } R * \text{conj}(R) / |R|^2)$ , not the filter's nominal centre frequency. Using the nominal value biases every octave differently.

## The hard limit

Phase is unique only in  $(-\pi, \pi]$ . Wrapping occurs once  $|\Delta\phi| > \pi$ , i.e.

$$\delta_{\text{max}} = \frac{1}{2k} = \frac{\lambda}{2} \text{ per sub-band}$$

Scaling is the worst case: displacement grows with radius,  $\delta(r) = (s-1)r$ . It is tiny at the centre (works) and large at the periphery (aliases) — exactly the mixed failure seen below.

## Two stabilising ideas used in both tools

- Singularity rejection: where amplitude  $\rho = |R| \rightarrow 0$  the phase is undefined and its gradient diverges; such pixels are masked (Fleet & Jepson, Stability of Phase).
- Aperture problem: per-orientation component displacements are fused into a full 2D (U,V) vector by weighted least squares over all orientations.

## Resolution (multi-scale)

A Gaussian pyramid decomposes a large motion into small per-level sub-motions: at level  $L$  the motion is downsampled by  $2^L$  until it fits under  $\lambda/2$ . The estimate is upsampled, used to rewrap the next finer level, and only the small residual is measured there (no unwrapping). Summing across levels reconstructs the full field.

## Single-scale Fleet-Jepson phase flow

One octave band per orientation, measured local frequency as denominator, amplitude + frequency-consistency masking, least-squares fusion over orientations, coarse-to-fine only within one FFT (no image pyramid). Confidence-weighted smoothing propagates edge measurements into flat regions for display.

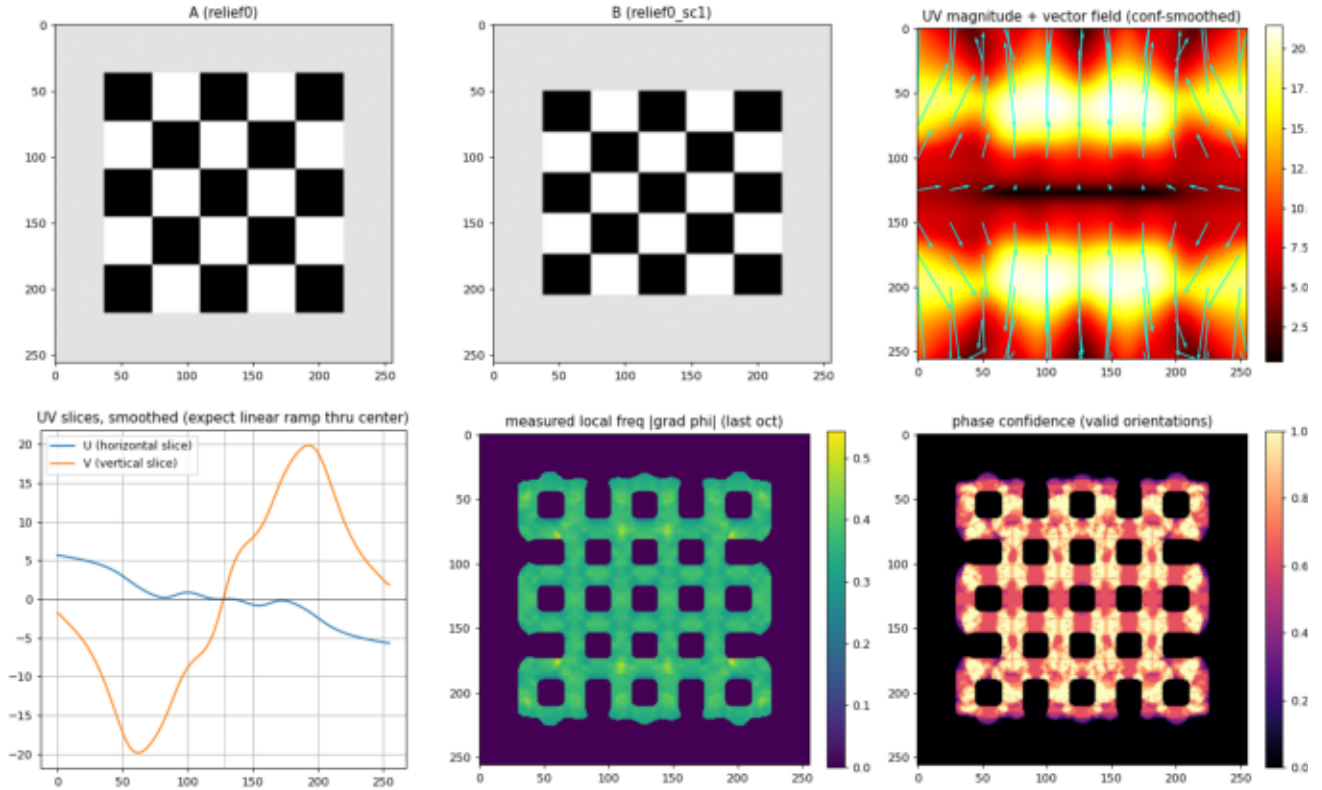


Fig. A1 A, B (checkerboards); UV magnitude + vector field; UV centre slices; measured local frequency; phase confidence.

## Observations

- Local-frequency map is clean ( $\sim 1.0$ ) and the confidence map correctly lights up the edges — the Fleet-Jepson core works.
- Auto-selected octaves 2-4 (checkerboard fundamental); higher bands carry no energy.
- Displacement max  $\sim 21$  px, mean  $\sim 11$  px; estimated divergence  $\rightarrow s \sim 1.16$ .
- UV slices are chaotic away from the centre: the peripheral displacement exceeds  $\lambda/2$  and aliases. This is the documented failure mode, not an implementation bug.

## Multi-scale coarse-to-fine (M-PME)

Gaussian pyramid (5 levels), coarse-to-fine with rewarping and per-level residual refinement, no phase unwrapping, motion-field fusion across scales. Reimplementation of the method described in 'Large amplitude motion estimation via multi-scale phase-based video processing' (Mech. Syst. Signal Process. 2026), not the original paywalled code.

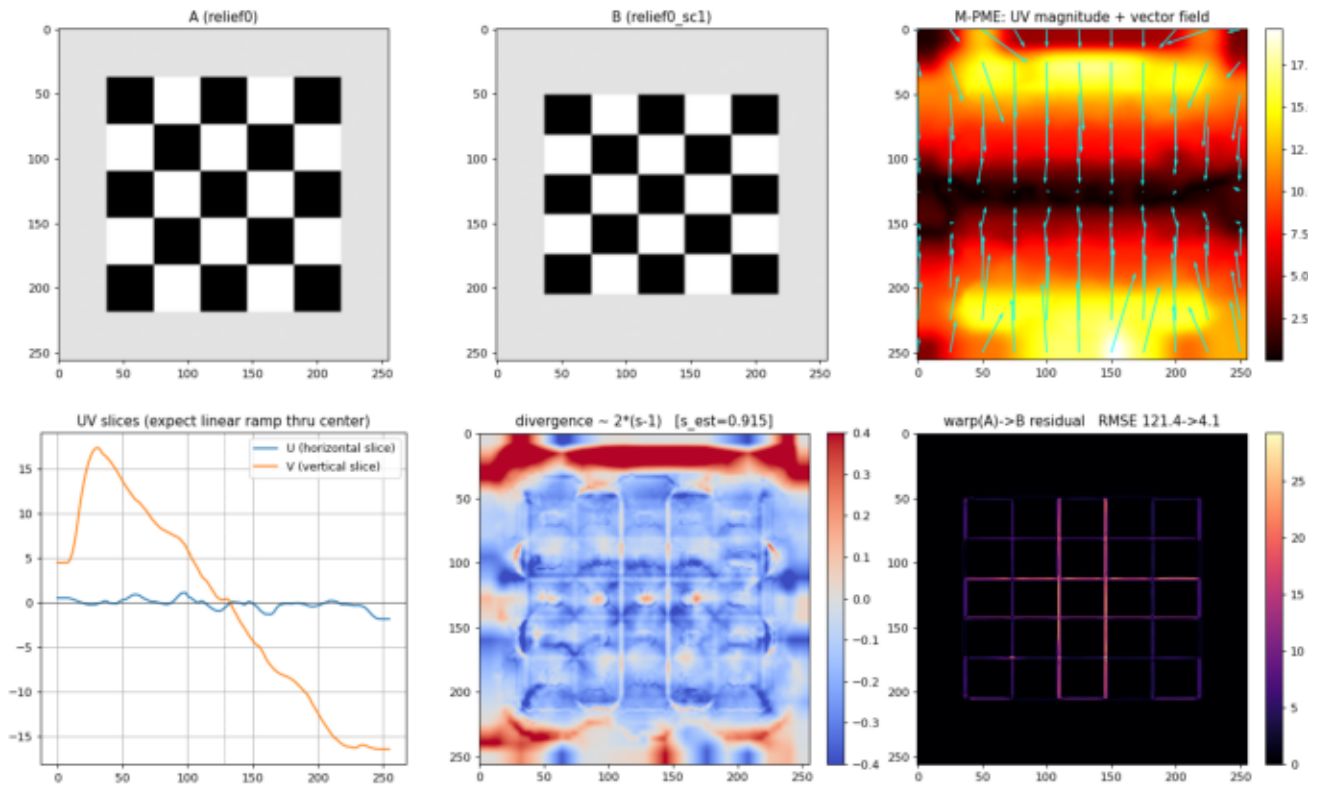


Fig. B1 A, B; UV magnitude + vector field; UV centre slices; divergence (local scale); warp(A)->B residual.

## Results

| Quantity                | Value                            |         |
|-------------------------|----------------------------------|---------|
| Warp residual RMSE (fg) | 121.4 -> 4.11 px                 | (-97%)  |
| mean flow  per level    | 0.55 / 1.10 / 2.19 / 4.38 / 8.76 | doubles |
| max / mean displacement | 19.6 / 8.8 px                    |         |

- Converges: warp(A, flow) reconstructs B to ~4 px RMSE (thin edge residual only).
- mean|flow| doubles cleanly per level — the expected coarse-to-fine signature.
- Field is strongly vertical-dominant (V slice +17 px, U slice +1 px). Since the warp reconstructs B, this is real: relief0\_sc1 is NOT an isotropic scale of relief0 but anisotropic (vertical scale/shift). A single scalar  $s \sim 0.915$  is therefore misleading.

## Benchmark summary & next steps

| Aspect        | Single-scale (A)         | Multi-scale (B)        |
|---------------|--------------------------|------------------------|
| Large motion  | aliases ( $>\lambda/2$ ) | handled                |
| Warp residual | not usable               | 121 $\rightarrow$ 4 px |
| UV slices     | chaotic                  | approx. linear ramp    |
| Cost          | 1 FFT set                | pyramid x refine       |

## Takeaways

- The pyramid argument holds: coarse-to-fine contextualisation overcomes the  $\lambda/2$  ceiling, because the wrap limit only ever applies to the small per-level residual.
- Sign matters: from  $d\phi = -k \cdot \delta$ , the displacement is  $-d\phi/k$ ; a wrong sign makes the warp diverge (residual grew to 175 px before the fix).
- The test data is anisotropic — clarify what `relief0_sc1` actually is before using it as a scalar-scale ground truth.

## Why this is still not enough for 2D- $\rightarrow$ 3D

The 4 px residual is global reconstruction energy, not per-pixel depth accuracy. Phase flow is defined only on textured edges and interpolated elsewhere, so the field lacks the sub-pixel smoothness a depth reconstruction needs. Closing that gap is the motivation for ImplicitKalman.

## References

Fleet & Jepson 1990, Computation of component image velocity from local phase information, IJCV 5(1).  
Fleet & Jepson 1993, Stability of phase information, IEEE PAMI 15(12).  
Barron, Fleet & Beauchemin 1994, Performance of optical flow techniques, IJCV 12(1).  
Wadhwa et al. 2013, Phase-based video motion processing, ACM TOG (SIGGRAPH).  
Large amplitude motion estimation via multi-scale phase-based video processing, Mech. Syst. Signal Process. 2026 (S0888327026004589).